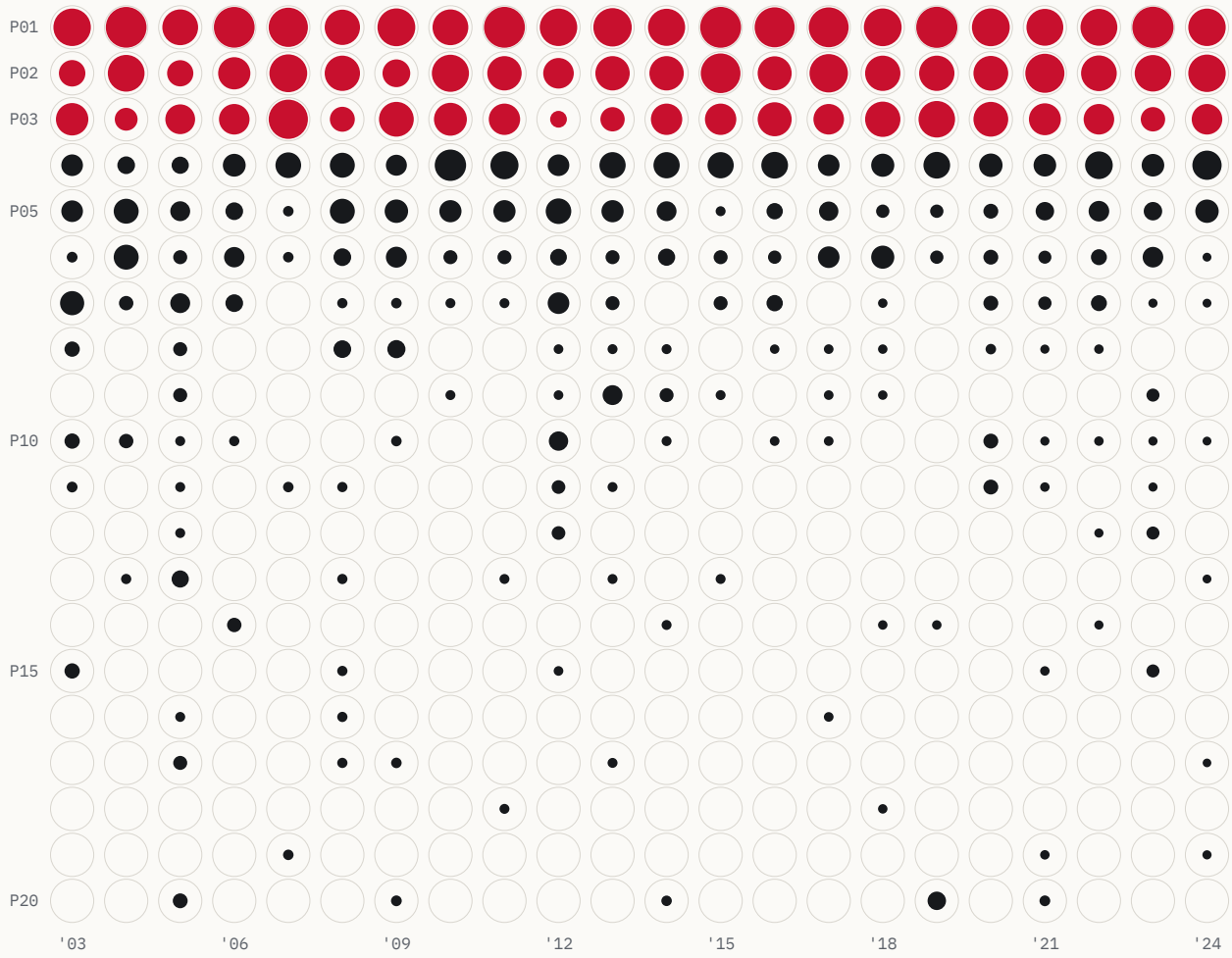
Predicting the Formula One podium *before the lights go out.*

A leakage-safe pipeline from fourteen relational tables to a podium probability for every driver, using only what is known after qualifying and before the start.



FIELD 01 PODIUM SHARE BY STARTING SLOT, 2003–2024. AREA = SHARE OF RACES FINISHED IN THE TOP THREE.

TEAM

Tony Hany Omar Medhat Mina Magdy Maged Emil

00

Summary

0.708	0.647	8,970	18
TEST PR-AUC FINAL MODEL	TEST PR-AUC GRID-ONLY RULE	DRIVER-RACE ROWS 2003–2024	PRE-RACE FEATURES

We converted the 14 relational Kaggle/Ergast Formula 1 tables (1950–2024) into a validated, leakage-safe modelling table with one row per driver per race, answered three data-engineering questions, and trained three models to predict whether a driver finishes on the podium using only information available after qualifying and before the start. On the unseen test seasons 2022–2024 all three models beat a grid-only rule (PR-AUC 0.70–0.71 vs 0.65), and they are practically tied with each other. The final model is a shallow neural network (test PR-AUC 0.708, ROC-AUC 0.938). Starting position and qualifying position dominate the explanations; they largely replace each other, but removing both costs about 0.14 PR-AUC.

ITEM	RESULT
Modelling table	8,970 rows (one per driver per race), 2003–2024, 18 pre-race features, podium rate 14.3%
Split (by season)	train 2003–2019 · validation 2020–2021 · test 2022–2024
Models	logistic regression (baseline), random forest, shallow neural network
Primary / secondary metric	PR-AUC / ROC-AUC
Final model, test	**neural network: PR-AUC 0.708, ROC-AUC 0.938, F1 0.669 (recall 0.85, precision 0.55)**
Q1	No detectable change in the circuit ranking after 2014; front-row conversion rose from 68% to 76%
Q2	No home advantage once starting position is controlled (–1.5 points, 95% CI –4.5 to +1.2)
Q3	Highest mechanical-retirement rates: Lotus F1 and Toro Rosso (2014–2021); Alfa Romeo and Alpine (2022–2024)

01

Task and data

Task. Binary classification: does a driver finish on the podium (positionOrder ≤ 3)? **Unit of analysis:** one driver entering one race, so (raceId, driverId) must be unique. **Prediction cutoff:** after qualifying, before the start. Results, status, points, laps and pit stops of the current race, and the standings published after it, are never features.

Data. Formula 1 World Championship (1950–2024), 14 linked CSV tables. The raw files were never modified: an MD5 fingerprint of every file is taken at the start of the notebook and verified at the end.

Modelling window: 2003–2024. This is our choice, made from the data. The prediction cutoff makes qualifying a core input, and the audit showed that qualifying records do not exist before 1994 and cover only 6–58% of entries in 1996–2002, but are complete from 2003 (Figure 1). Training on years where qualifying is mostly missing would teach the model about gaps in the data rather than about racing.

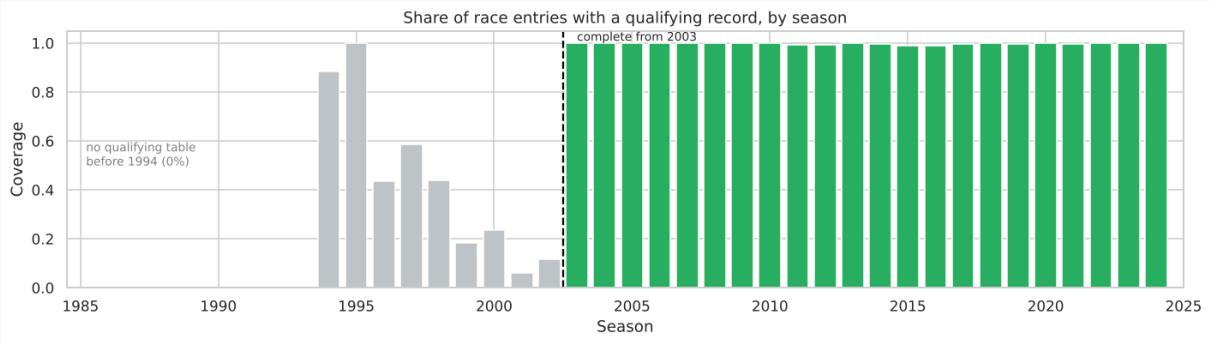


FIG 01 Share of race entries with a qualifying record, by season.

02

Audit and cleaning

Every problem was found by code and plots before anything was changed; every cleaning step works on copies and ends with a check.

ISSUE FOUND IN THE AUDIT	CLEANING RULE	CHECK
Placeholders: \N in 30 columns, empty strings in 68 qualifying values	Both → missing (NaN); no filling at this stage	NaN count per column equals the placeholder count; 160,144 cells
Everything stored as text; lap times as “1:26.572”	Numbers, dates and seconds, by measured parse rates	87 columns typed, 0 values lost
85 driver-race pairs appear twice (1950s shared drives)	Features from the car the driver started in; podium = 1 if any of his rows finished top 3	(raceId, driverId) unique; e.g. Moss, 1960 Argentine GP
grid = 0 (pit-lane start or no start)	grid → missing + flag no_grid_slot; rows kept	86 entries since 2003, 0 podiums
One Q1 time 10× the pole time (1995)	Set to missing (rule: > 3× race-best)	Next slowest is 1.46×
Primary and foreign keys	—	All 14 primary keys valid; all 23 foreign keys reconcile, before and after cleaning

Choosing the duplicate row by its finishing position would use post-race information; keeping the starting car is consistent with the pre-race cutoff, and the target still records the driver's podium. No duplicate falls inside 2003–2024, so the rule does not affect the model.

Data-engineering questions

All three answers use multi-hop joins onto results: races (season, circuit), circuits (country), qualifying (qualifying position), drivers (nationality), constructors (team) and status (cause of retirement). The joins are left joins, and an assertion confirms that the joined table still has exactly one row per driver per race (26,668 rows).

3.1 Q1 — Which circuits convert a front-row start into a podium most often, and does that ranking change after 2014?

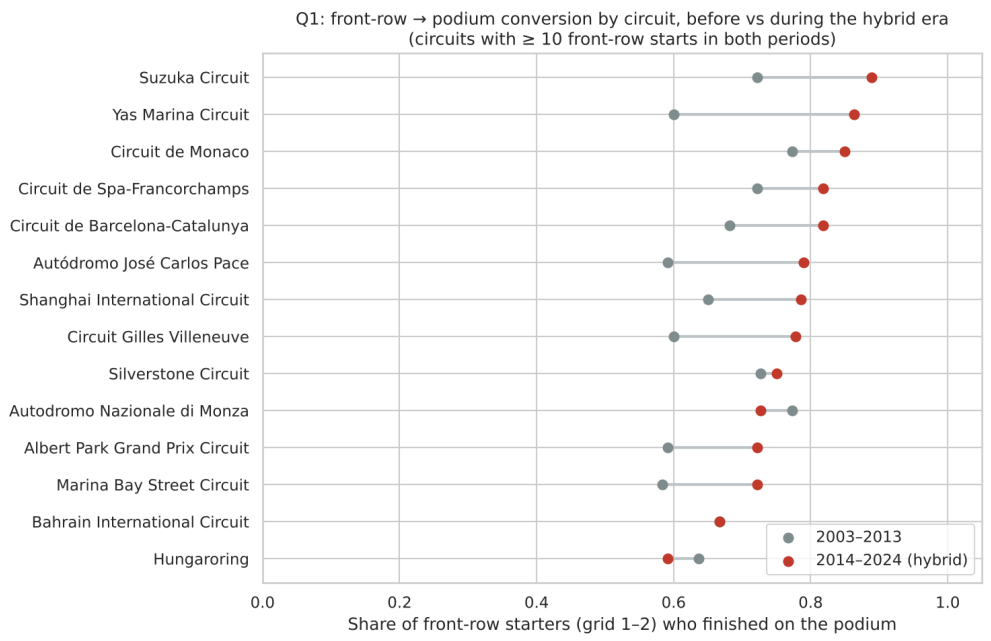


FIG 02 Front-row (grid 1–2) conversion per circuit, 2003–2013 (grey) vs 2014–2024 (red), circuits with at least 10 front-row starts in both periods.

Answer. The highest conversion rates were Magny-Cours (92%, 12 starts), Istanbul Park (79%) and Monza and Monaco (77%) before 2014, and Suzuka (89%), Yas Marina (86%) and Monaco (85%) from 2014. The order looks reshuffled (rank correlation 0.30, p = 0.29), but a chi-square test finds no real difference between circuits within either period (p = 0.95 and 0.74; a simulation that does not rely on the chi-square approximation gives 0.95 and 0.75). With 10–26 front-row starts per circuit, chance alone produces a reshuffle this large, so **the data cannot show that the ranking changed**. What clearly changed is the level: front-row conversion rose from 68% to 76% (z = 2.6, p ≈ 0.01) and rose at 11 of 14 circuits. Measured by qualifying position instead of grid, conversion is slightly lower (67% / 74%): 38 front-row qualifiers started further back after penalties and only 11 still reached the podium; the circuit ranking barely changes (rank correlation 0.94 / 0.88).

Grain: driver-race rows 2003–2024, filtered to front-row starters, aggregated to circuit × period. **Missingness:** entries without a grid slot or qualifying record cannot be front-row starters under that definition; the target is never missing. **Cutoff:** grid and qualifying are pre-race; the podium is the outcome. A minimum of 10 front-row starts per period keeps luck out of the ranking; every rate has a 95% Wilson interval.

3.2 Q2 — Do drivers podium more often at home, after controlling for starting position?

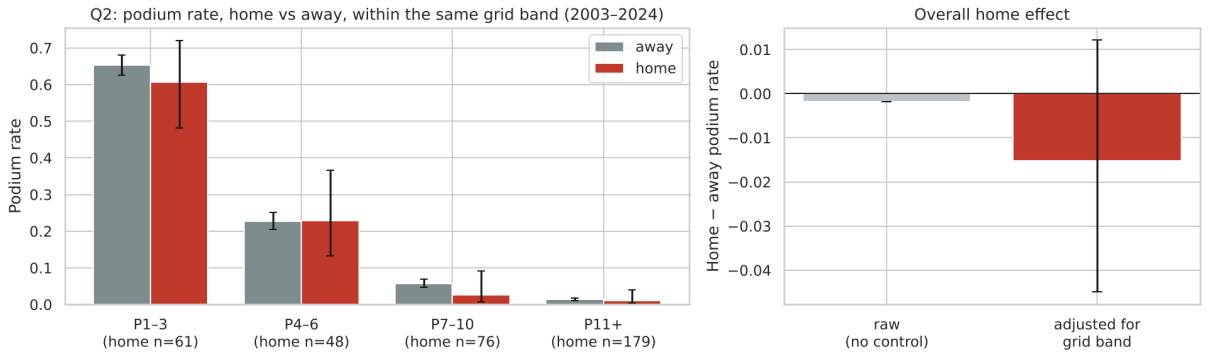


FIG 03 Podium rate at home vs away within the same grid band (2003–2024), and the overall home effect before and after controlling for grid.

Answer: no. A driver is at home when his nationality maps to the circuit's country (an explicit mapping; dual nationalities count in both countries). Comparing home and away drivers inside the same grid band (P1–3, P4–6, P7–10, P11+) and combining the bands weighted by home entries, the home effect is **−1.5 percentage points (95% bootstrap interval −4.5 to +1.2)**: zero lies inside the interval. Using all seasons since 1950 (Indianapolis 500 excluded) gives −0.4 (−1.6 to +0.8). Individual drivers differ: Hamilton (79% vs 56%) and Alonso (36% vs 27%) did better at home, Button never reached a home podium (0 of 14). Without these three, the rest do slightly worse at home (−3.3, −6.0 to −0.7), which we report as a pattern, not as a proven disadvantage.

Grain: driver-race rows, compared inside grid bands. **Missingness:** entries without a grid slot cannot be banded and are excluded; nationalities whose country never hosted a race are “away” by definition. **Cutoff:** nationality and circuit are known in advance, so a home flag is a legal feature. **Limitation:** only 364 home entries in 2003–2024, so a small effect could be hidden.

3.3 Q3 — Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

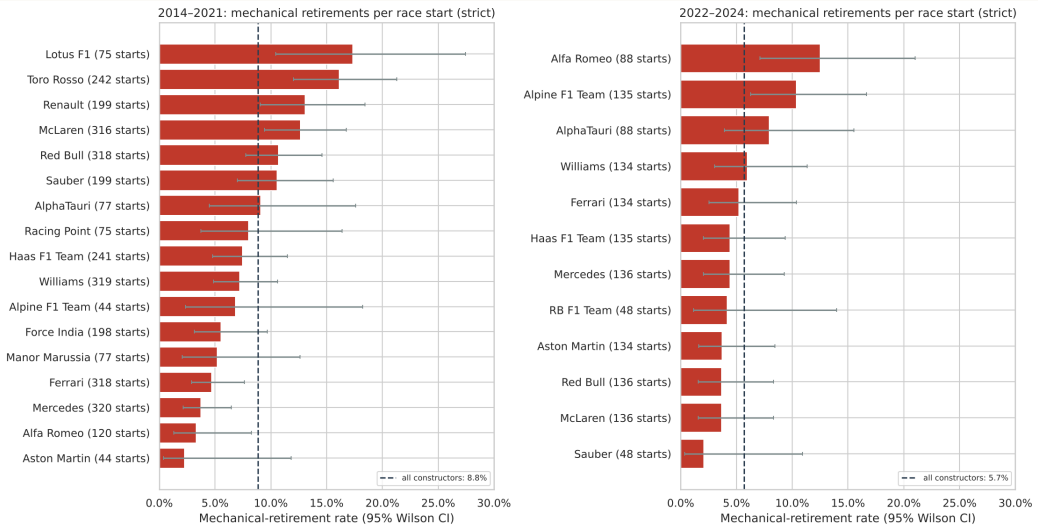


FIG 04 Strict mechanical retirements per race start, by constructor and era (95% Wilson intervals; constructors with at least 40 starts).

Definition (from the brief): a race start that ended early because a component of the car failed, excluding crashes, driver causes and regulation decisions. All 139 status codes are mapped to one

group. **Strict** counts statuses that name a component (engine, gearbox, brakes…); **broad** adds statuses with no clear cause (Retired, Withdrew after racing, Wheel, Puncture, Front wing…). Only race starts count: a car marked “W” in positionText never started (e.g. Leclerc, Monaco 2021, driveshaft failure on the way to the grid), so it is neither a start nor a retirement.

Answer. 2014–2021: **Lotus F1 (17.3% of 75 starts), Toro Rosso (16.1% of 242), Renault (13.1% of 199)**; Mercedes (3.8%) and Ferrari (4.7%) failed least among full-period teams. 2022–2024: **Alfa Romeo (12.5% of 88), Alpine (10.4% of 135), AlphaTauri (8.0% of 88)**. Across all teams the strict rate fell from 8.8% to 5.7% of starts (z = 3.60, p = 0.0003), although the decline began before 2022 (13.5% in 2017, 4.3% in 2021) and the first season of the new cars spiked to 8.9%. The broad definition keeps the same leader in each era.

Grain: race starts 2014–2024, aggregated to constructor × era. **Missingness:** status is never missing; the missing cause behind ambiguous statuses is handled by the strict/broad pair. **Cutoff:** status is post-race and is never a feature for the same race; the model only uses past reliability. **Caveat:** renamed teams have new IDs (Toro Rosso → AlphaTauri → RB; Lotus F1 → Renault → Alpine).

04

Features and why each is valid at the pre-race cutoff

Rows are ordered by (year, round), never by raceId, which is not in date order (raceId 1 is the 2009 Australian GP). Form features shift by one race before rolling, so race r only sees races before r. Standings are taken from after the previous round, because the standings table for race r already contains race r's points.

GROUP	FEATURE	WHY IT IS KNOWN BEFORE THE RACE
Grid	grid; no_grid_slot; field_size	The grid is fixed before the start; the entry list is published before the weekend
Qualifying	quali_pos; gap_to_pole_pct; grid_minus_quali	Qualifying is over; grid − qualifying shows penalties
Driver form	avg finish and podium rate over the last 5 races; prior races	Earlier races only (shift(1) before rolling)
Constructor form	team avg finish and podium rate (last 5); mechanical-failure rate (last 10)	Earlier races only, one row per team per race
Standings	driver and team position; points share of the leader	Published after the previous round; round 1 empty / 0
Context	home_race; round	Calendar and nationality

Evidence. All feature evidence uses the training seasons only (podium rate by feature value, univariate AUC, correlations, missingness). Qualifying position (AUC 0.910), grid (0.909), gap to pole (0.893) and championship position (0.893) are strongest alone; all 18 were kept and their contribution was measured afterwards by ablation (Section 6). Points share is used instead of raw points because the scoring system changed in 2010; qualifying position and gap instead of “reached Q3”, because Q2/Q3 do not exist before 2006.

Leakage controls. Every column of the joined table is labelled pre- or post-race and post-race columns are blocked by an assertion. A perturbation test scrambles one race's result (2015 Hungarian GP): that race's features stay identical, while the next race's form and standings features change, which proves the test can detect leakage. Experiment E3 shows why this matters: letting form include the current race inflates validation PR-AUC from 0.710 to 0.847,

while using post-race standings changes it by only −0.007, so a score alone can never prove the absence of leakage.

Split and preprocessing. A split by season mirrors real use and keeps future seasons out of training: train 2003–2019 (6,831 rows), validation 2020–2021 (780), test 2022–2024 (1,359; also the first seasons of new car rules). Same-race domain fills happen before the split (no grid slot → back of the grid; no lap time → slowest gap of that race; round 1 → 0 points). Statistical fills (training median plus a missing-indicator column) and scaling are fitted inside the model pipeline on training data only. Fitting them on all seasons (E1) or scaling before imputing (E2) changed validation PR-AUC by less than 0.001, so the correct order costs nothing.

05

Models and results

Models. Logistic regression (a weighted sum of features turned into a probability; the baseline), a random forest (300 decision trees on random samples, averaged) and a shallow neural network (two hidden layers of 64 and 32 ReLU neurons, Adam, L2 penalty), all on the same rows, features and split. **Primary metric: PR-AUC**, because podiums are only 15% of rows and PR-AUC focuses on ranking them correctly; random guessing scores ≈ 0.15. **Secondary: ROC-AUC.** F1, precision, recall and accuracy are also reported; accuracy never decides, since always predicting “no podium” is already 85% accurate.

Tuning without validation or test. Hyperparameters were chosen by season-based cross-validation inside 2003–2019 (train on earlier seasons, score 2015, 2016, … 2019). The network's first attempt used scikit-learn's early stopping on accuracy, which stops too early with 15% positives (CV PR-AUC 0.720); without it and with a stronger penalty it reached 0.748. The decision rule was chosen on validation: a tuned threshold (mean F1 0.706) beat “top three probabilities per race” (0.684). The final model was the highest validation PR-AUC, keeping the simpler model if two are within 0.01 — fixed before the test set was used.

MODEL (TEST 2022–2024)	PR-AUC	ROC-AUC	F1	PRECISION	RECALL	ACCURACY
Reference: grid ≤ 3 (no learning)	0.647	0.914	0.649	0.650	0.647	0.895
Logistic regression (baseline)	0.707	0.936	0.671	0.559	0.838	0.876
Random forest	0.700	0.938	0.674	0.630	0.725	0.895
Neural network (final)	**0.708**	**0.938**	**0.669**	**0.553**	**0.848**	**0.874**

Validation PR-AUC: logistic regression 0.720, random forest 0.719, neural network 0.732 (final model).

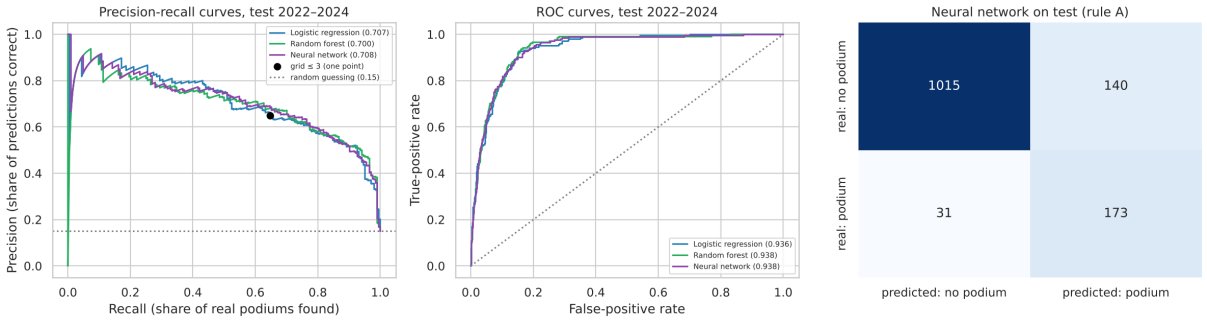


FIG 05 Test set: precision-recall curves, ROC curves and the final model's confusion matrix (173 of 204 podiums found, 140 false alarms).

Interpretation. All three models beat the grid rule on every split, so the features beyond the grid add real information. They are practically tied with each other (differences below 0.01 on 204 test podiums), so the features matter more than the model; the simple baseline is almost as good as the network. The random forest overfits somewhat (training PR-AUC 0.774 vs validation 0.719); logistic regression and the network do not, and the network's validation score rises steadily during training and then stays flat. By test season, 2023 (Red Bull won 21 of 22 races) was easiest (≈ 0.78) and 2024 (four winning teams) hardest; in 2022, the first season of the new rules, all models still beat the grid rule.

06

Experiments and explainability

6.1 Feature-group ablation

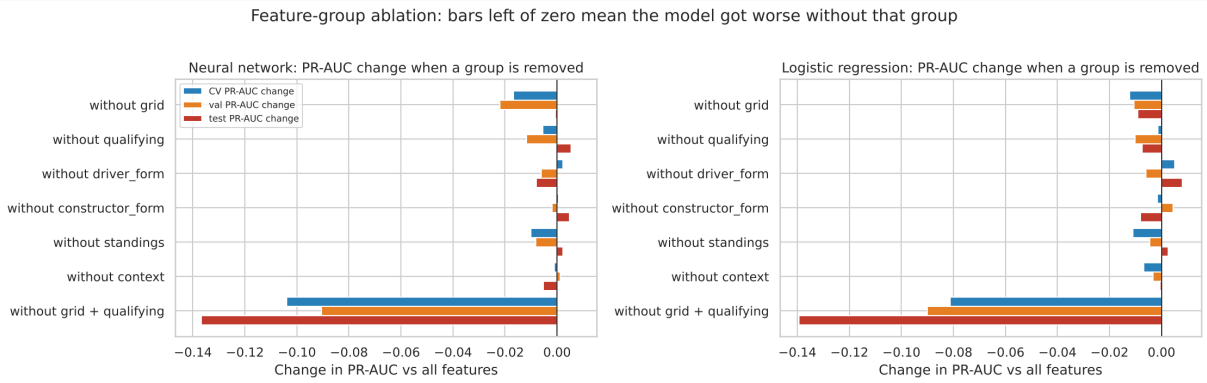


FIG 06 Change in PR-AUC when each feature group is removed and the model refitted (blue = CV, orange = validation, red = test).

Because grid_minus_quali equals grid minus qualifying position, the “without grid” variant also drops it; otherwise the model could rebuild grid exactly from the qualifying group. Removing **grid** costs the network 0.017 (CV), 0.022 (validation) and 0.001 (test); removing **qualifying** costs 0.006 and 0.012 (on test it gains 0.006). Removing **both together** costs 0.104, 0.091 and 0.137, and similarly for logistic regression: each group can stand in for most of the other (correlation 0.97), and together they are by far the most important information. **Grid** is the most useful single group in cross-validation, slightly ahead of **standings** (−0.010). Driver form, constructor form and context change PR-AUC by less than 0.01 in either direction, which is within the noise between CV, validation and test; their information is largely covered by grid, qualifying and standings. Both models show the same pattern.

6.2 Global explanations: permutation importance and SHAP

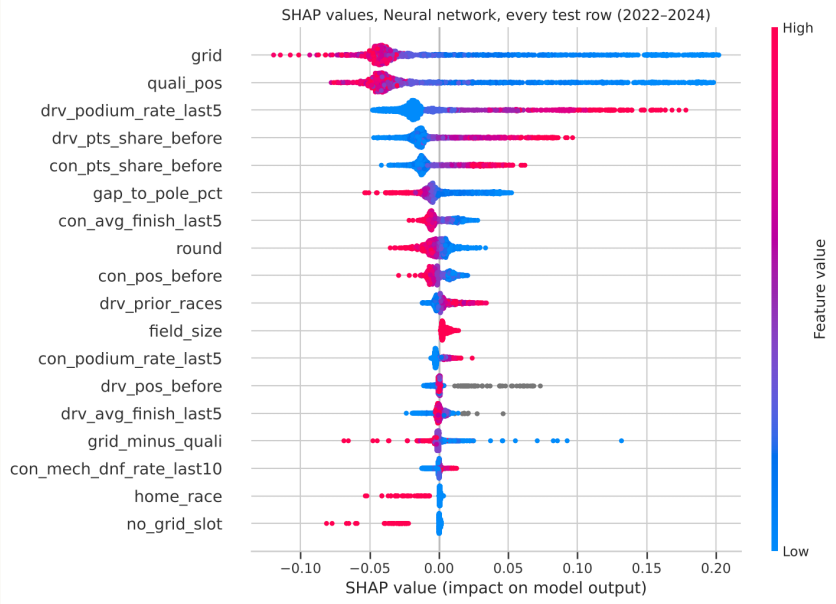


FIG 07 SHAP values of the final model for every test row (red = high feature value, blue = low; right = pushes the podium probability up).

Permutation importance (test PR-AUC drop when one feature is shuffled, 20 repeats): grid (0.124) and qualifying position (0.122) dominate, followed by the driver's recent podium rate (0.028), points share (0.014) and gap to pole (0.013); all others are below 0.004. **SHAP** (model-agnostic permutation explainer, 100 background rows; the values add up exactly to each prediction) agrees on the top four: grid, qualifying position, recent podium rate and points share. A front-row start adds about +0.15 to +0.18 to the probability, while from about P8 onward the effect is flat; a good grid slot counts more when the team has recently been on the podium, an interaction a linear model cannot learn. Permutation and ablation look different for grid because they answer different questions: shuffling grid breaks a model trained to use it, while retraining without grid lets the model use qualifying instead.

6.3 Local explanations: SHAP waterfall and LIME

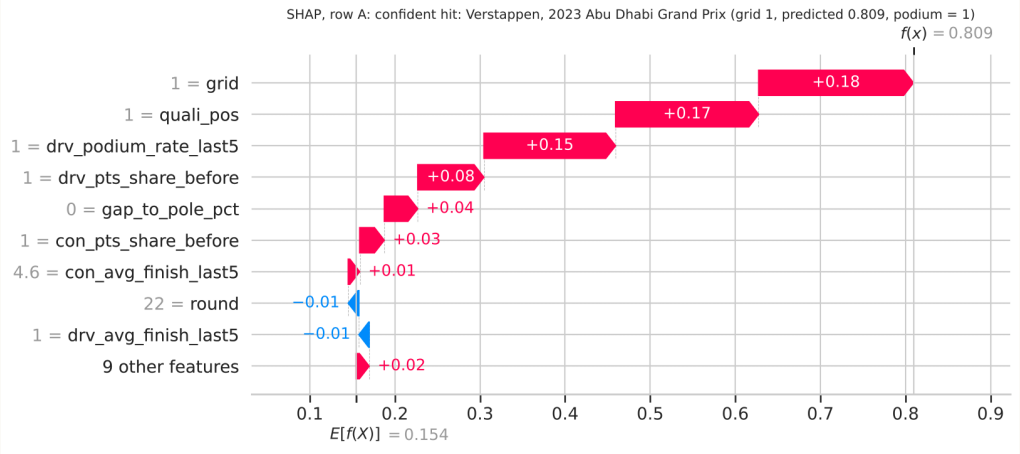


FIG 08 SHAP waterfall for one prediction: Verstappen, 2023 Abu Dhabi GP, from pole (predicted 0.81; podium).

For Verstappen at the 2023 Abu Dhabi GP, pole position (+0.18), qualifying first (+0.17), five recent podiums (+0.15) and leading the championship (+0.08) lift the probability from the average 0.15 to 0.81. His average finish of 1 gets a small negative value (−0.01): it overlaps with the podium rate

and standings, which take the credit, and changing only that value moves the probability by less than 0.02, so it is a side effect of correlated features, not an error. The biggest miss is Gasly at the 2024 São Paulo GP: grid 13, probability 0.003, yet third in a rain-hit race with a red flag, which no pre-race information can foresee. LIME (a local linear model fitted to 5,000 perturbed copies of the row) shares 4 of its top 5 features with SHAP for both rows; it imitates the network well for the first row (R^2 0.43) and poorly for the second (R^2 0.16), so SHAP is used for exact values and LIME as a cross-check.

> NOT CAUSAL
>
> **These explanations are not causal effects.** A large SHAP value or permutation importance for grid means the model relies on starting position to make its prediction. It does not prove that starting position causes podium finishes: a fast car produces both a good grid slot and a good result. Likewise, a high points share does not cause the next podium; it reflects a strong driver and car. Showing cause would require a controlled experiment, which observational race data cannot provide.

6.4 Inference

The function predict_podium() takes one driver's pre-race values, runs them through the final pipeline and returns the probability, the decision at the validation threshold (0.26) and the list of filled fields. For Norris on pole at the 2024 Abu Dhabi GP it returns 0.708 (podium; he won). With the qualifying and form fields removed it still works: the gaps are filled with training medians and the probability falls to 0.508, still a podium. Misspelled field names are rejected instead of being silently treated as missing.

07

Known limitations

- **Qualifying-coverage drift:** qualifying data is missing for most entries before 2003, so the model only covers 2003–2024 and cannot be applied to earlier eras.
- **Era sensitivity:** rules, scoring systems and field sizes change between seasons. The 2022 rule change is in the test set and the models still beat the grid rule there, but 2024, with four winning teams, was the hardest season: form built on past races helps less when the order changes during a season.
- **Recording conventions:** in 2003 grid equals qualifying position for almost every entry (and only 6% differ in 2007), which most likely reflects how the source recorded those seasons; missing grid slots are much more frequent since 2019.
- **Renamed teams** get new IDs, so their first races under a new name have no team form (filled and flagged).
- **Small samples:** validation has 117 podiums, Q1 and Q3 rest on 10–26 front-row starts per circuit and 40–320 starts per team, and Q2 on 364 home entries; small differences are within noise.
- **Missing qualifying at prediction time** is filled with a training median without an indicator, because qualifying was never missing in training; the inference function reports which fields were filled.
- **Independent predictions:** each driver is scored separately, so the probabilities of one race do not add up to three podiums; unforeseeable race events (weather, crashes, safety cars) limit any pre-race model.

08

Conclusion

A leakage-safe table of pre-race information predicts podium finishes well (test PR-AUC 0.71, ROC-AUC 0.94), and clearly better than starting position alone. Most of the signal comes from where a driver starts and how fast he qualified, with recent form and championship standing adding the rest. The choice of model matters little: a regularised logistic regression is almost as good as the neural network.